

DHAKA INTERNATIONAL UNIVERSITY

Report Name

Testing and Validation Report of the Outfleck E-Commerce Platform

A REPORT SUBMITTED TO

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Repository: **github.com/vampxlr/outfleck**

Live Application: **outfleck.vercel.app**

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1. Introduction

Outfleck is a Bangladesh-market t-shirt e-commerce platform built as a full production web application. It exposes three surfaces:

1. A **storefront** (home/listing, product detail, cart, checkout, order tracking) that Bangladeshi shoppers use to browse products and place orders, with prices shown in BDT (৳) and payment via Cash on Delivery (COD) or bKash mobile money.
2. **Ad landing pages** (`/l/<slug>`) — single-purpose, marketer-configured pages used to drive paid-traffic conversions with overridden pricing, custom delivery fees, and an inline order form.
3. An **admin dashboard** for order management, product/catalogue management, landing page building, and store settings (including Meta Conversions API / CAPI tracking configuration).

Technology Stack

LAYER	TECHNOLOGY
Framework	Next.js 15 (App Router)
Language	TypeScript
Backend / database	Convex (reactive serverless functions + document DB)
Animation	Framer Motion
Styling	Tailwind CSS v4
Client state	Zustand
Hosting	Vercel

Repository: <https://github.com/vampxlr/outfleck> **Live Application:** <https://outfleck.vercel.app>

Scope of this Report

This report documents the functional (black box) testing of all customer-facing and admin surfaces of Outfleck, and the structural (white box) testing of the pure business logic extracted into `lib/pricing.ts` and `lib/validation.ts`. All results recorded here were produced by executing the Vitest unit suite and a set of scripted black box/API probes against the running Convex development deployment — none of the results in this report are hypothetical.

2. Test Environment

ITEM	DETAIL
Operating System	Windows 11 Pro
Node.js	v24.14.1
Application framework	Next.js 15.5.22
Backend deployment	Convex dev deployment <code>handsome-hornet-178</code>
Browser (E2E/UI)	Chromium, driven via Playwright
Unit test runner	Vitest 4, with <code>@vitest/coverage-v8</code>
Test execution date	19 August 2026

3. Testing Methodology

Two complementary testing philosophies were applied.

Black Box Testing evaluates the system purely from the outside — inputs are supplied and outputs/behaviour are observed, without reference to the internal source code. It is the natural approach for a system's user-facing pages and public APIs, because it verifies that the software meets its functional requirements from the end user's point of view. Its main advantage is that it does not require the tester to understand the implementation and it exercises the system the way a real user or attacker would; its main limitation is that it cannot guarantee that every internal code path has actually been exercised, since two very different implementations can produce identical external behaviour. Within black box testing this report uses:

- **Equivalence partitioning** — grouping inputs (e.g. valid vs invalid phone numbers) into classes expected to be handled the same way, and testing one representative from each class.
- **Boundary value analysis** — testing values exactly at, just below, and just above a threshold (e.g. the £2000 free-delivery cutoff), since defects cluster at boundaries.
- **Decision table testing** — enumerating combinations of conditions (e.g. payment method × field validity) and the action expected for each combination.
- **State transition testing** — verifying that an entity (an order) moves correctly through its defined states (pending → confirmed → shipped → delivered / cancelled).

White Box Testing evaluates the system with full knowledge of its internal source code and logic. It is the natural approach for the pure, side-effect-free business-logic functions in `lib/pricing.ts` and `lib/validation.ts`, because it lets the tester design inputs that deliberately exercise every statement, branch, and decision path inside the function. Its advantage is that it gives a quantifiable measure of how thoroughly the code was exercised (coverage percentages, independent path counts); its limitation is that it requires the tester to read and understand the implementation, and coverage alone does not guarantee the absence of *logical* (specification) defects. Within white box testing this report uses:

- **Statement coverage** — the percentage of executable statements run by the suite.
- **Branch coverage** — the percentage of `if` / `||` / `&&` /ternary branches taken in both directions.
- **Basis path testing** — deriving the minimum set of independent paths through a function from its control-flow graph, using McCabe's cyclomatic complexity.
- **Cyclomatic complexity** — a count of a function's linearly independent paths, computed as decision points + 1, used to estimate how many test cases a function needs.
- **Code walkthrough** — a manual, line-by-line read-through of a critical function (the `orders.place` mutation) to verify its logic and side effects.

Additionally, **Security Testing** (verifying that privileged data and operations cannot be accessed or forged by an unauthenticated/unauthorized client) was applied to the admin authentication surface, the CAPI token setting, and server-side price recomputation.

3.1 Home / Product Listing Page

Home Page Loads Product Grid

PASS

TEST CASE ID

TC HOME 01

TEST SCENARIO

Verify the storefront home page loads and lists all active products with prices

PRE-CONDITION

User navigates to the Outfleck storefront root URL

TEST STEPS

1. Open the home page in the browser.
2. Wait for the product grid to render.
3. Count the number of product cards and verify each shows a price in ₹.

TEST DATA

- URL: <https://outfleck.vercel.app/>

EXPECTED RESULT

- All active products should be displayed in a grid
- Each product card should show name, image, and price in BDT (₹)

ACTUAL RESULT

The home page rendered a grid of 7 products, each with a thumbnail image, product name, and price prefixed with ₹.

As shown in Figure 1, the home page displays the full product listing grid.

Figure 1: Home Page — Product Listing

Category Filter Narrows the Grid

PASS

TEST CASE ID

TC HOME 02

TEST SCENARIO

Verify selecting a category filter reduces the visible product set

PRE-CONDITION

Home page is loaded with the full product grid visible

TEST STEPS

1. Click a category filter chip/tab (e.g. "Graphic Tees").
2. Observe the product grid.

TEST DATA

- Category: Graphic Tees

EXPECTED RESULT

- Only products belonging to the selected category should remain visible
- The product count should decrease from the full catalogue count

ACTUAL RESULT

Selecting the category filter re-rendered the grid with only the matching subset of products; the count dropped from 7 to the category's product count.

Responsive Layout on Mobile Viewport

PASS

TEST CASE ID

TC HOME 03

TEST SCENARIO

Verify the home page adapts to a small mobile viewport

PRE-CONDITION

Home page is loaded

TEST STEPS

1. Resize/emulate the viewport to 390px width (iPhone-class device).
2. Observe the product grid layout and navigation.

TEST DATA

- Viewport width: 390px

EXPECTED RESULT

- Product grid should reflow to a mobile-appropriate column count
- Navigation/header should collapse to a mobile-friendly layout

ACTUAL RESULT

At 390px width the grid reflowed to a single/double-column mobile layout with no horizontal overflow, and the header collapsed correctly.

As shown in Figure 2, the home page renders correctly at a 390px mobile viewport.

Figure 2: Home Page — Mobile Viewport (390px)

Add to Cart From Listing Card

PASS

TEST CASE ID

TC HOME 04

TEST SCENARIO

Verify adding a product to the cart directly from a listing card opens the cart drawer

PRE-CONDITION

Home page is loaded with product grid visible

TEST STEPS

1. Click the "Add to Cart" control on a product card.
2. Observe whether the cart drawer opens.

TEST DATA

- Product: Noir Pinstripe Bloom Shirt, default size

EXPECTED RESULT

- The item should be added to the cart
- A cart drawer/panel should open showing the added item

ACTUAL RESULT

The item was added to the cart state and the cart drawer opened automatically, showing the newly added line item.

As shown in Figure 4, the cart drawer opens after adding an item from the listing page.

Figure 4: Cart Drawer After Add-to-Cart

Testing Approach for Home / Product Listing Module

The Home module is tested using Black Box Testing. Test cases exercise the page purely through user interactions — loading the page, applying filters, resizing the viewport, and clicking add-to-cart — and observe the resulting UI state without reference to component internals. Equivalence partitioning is used implicitly for the category filter (all products in a category are treated as one equivalence class), and the

mobile test is a black-box responsive-layout check. Since only externally observable behaviour is verified, this module is categorized under Black Box Testing.

3.2 Product Detail Page

Product Detail Renders Correctly

PASS

TEST CASE ID

TC PROD 01

TEST SCENARIO

Verify a product detail page renders the correct gallery, price, and description

PRE-CONDITION

User navigates to a specific product's detail URL from the listing page

TEST STEPS

1. Click a product card on the home page.
2. Observe the product detail page.

TEST DATA

- Product: Noir Pinstripe Bloom Shirt

EXPECTED RESULT

- Image gallery, product name, price, and description should match the selected product

ACTUAL RESULT

The detail page rendered the correct image gallery, name "Noir Pinstripe Bloom Shirt", price in ₺, and description matching the catalogue entry.

As shown in Figure 3, the product detail page shows the gallery, price, and description.

Figure 3: Product Detail Page

Size Selection

PASS

TEST CASE ID

TC PROD 02

TEST SCENARIO

Verify a user can select a size variant before ordering

PRE-CONDITION

Product detail page is loaded with multiple size options

TEST STEPS

1. Click a size option (e.g. "M").
2. Observe the selected-size indicator.

TEST DATA

- Size: M

EXPECTED RESULT

- The selected size should be visually highlighted and stored as the active selection

ACTUAL RESULT

Selecting "M" highlighted the size button and set it as the active variant used for stock/price lookups.

Quantity Stepper

PASS

TEST CASE ID

TC PROD 03

TEST SCENARIO

Verify the quantity stepper increases and decreases the order quantity

PRE-CONDITION

Product detail page is loaded, a size is selected

TEST STEPS

1. Click the "+" stepper control twice.
2. Click the "-" stepper control once.

TEST DATA

- Starting quantity: 1

EXPECTED RESULT

- Quantity should read 2 after two increments and one decrement

ACTUAL RESULT

Quantity stepper correctly incremented to 2, then to 3, then decremented to 2.

Order Now Routes to Checkout

PASS

TEST CASE ID

TC PROD 04

TEST SCENARIO

Verify the "Order Now" button routes directly to checkout with the selected item

PRE-CONDITION

Product detail page is loaded, a size and quantity are selected

TEST STEPS

1. Click the "Order Now" button.
2. Observe navigation.

TEST DATA

- Product: Noir Pinstripe Bloom Shirt, size M, qty 1

EXPECTED RESULT

- Browser should navigate to the checkout page with the selected item pre-loaded

ACTUAL RESULT

Clicking "Order Now" routed to the checkout page with the selected item present in the order summary.

Testing Approach for Product Detail Module

The Product Detail module is tested using Black Box Testing. Variant selection and quantity stepping are tested using equivalence partitioning (any valid size/quantity behaves the same) and boundary checks (minimum quantity 1). Since the tests validate only observable UI state and navigation outcomes, this module falls under Black Box Testing.

3.3 Shopping Cart Page

Cart Items Display With Correct Line Totals

PASS

TEST CASE ID

TC CART 01

TEST SCENARIO

Verify cart items and their line totals are displayed correctly

PRE-CONDITION

One or more items have been added to the cart

TEST STEPS

1. Navigate to the cart page.
2. Verify each line shows product name, size, unit price, quantity, and line total.

TEST DATA

- Item 1: Noir Pinstripe Bloom Shirt, size S, qty 1, unit price ₪1,490
- Item 2: Sky Doodle Blossom Shirt, size S, qty 1, unit price ₪1,490

EXPECTED RESULT

- Line total should equal unit price $\times$ quantity
- Cart subtotal should equal the sum of all line totals (₪2,980)
- Because ₪2,980 $\geq$ the ₪2,000 threshold, delivery should show as Free

ACTUAL RESULT

The cart page displayed 2 items, each with a line total of ₪1,490 ($1,490 \times 1$), a subtotal of ₪2,980 matching the sum of the line totals, and the delivery row showed “Free” with the “You’ve unlocked free delivery” banner.

As shown in Figure 5, the cart page lists items with correct line totals.

Figure 5: Shopping Cart Page

Quantity Update Recalculates Totals

PASS

TEST CASE ID

TC CART 02

TEST SCENARIO

Verify changing an item's quantity recalculates the line total and subtotal

PRE-CONDITION

Cart page is open with at least one item

TEST STEPS

1. Increase the quantity of an item from 1 to 2.
2. Observe the line total and subtotal.

TEST DATA

- Item: Noir Pinstripe Bloom Shirt, unit price ₺1490, qty 1 → 2

EXPECTED RESULT

- Line total should update to ₺2980; subtotal should update accordingly

ACTUAL RESULT

Increasing quantity to 2 updated the line total to ₺2980 and the subtotal reflected the change immediately.

Remove Item From Cart

PASS

TEST CASE ID

TC CART 03

TEST SCENARIO

Verify removing an item deletes it from the cart and updates totals

PRE-CONDITION

Cart page is open with at least one item

TEST STEPS

1. Click the remove/delete control on a cart line.
2. Observe the cart contents and subtotal.

TEST DATA

- Item: Noir Pinstripe Bloom Shirt

EXPECTED RESULT

- Item should be removed from the cart list
- Subtotal should decrease by the removed item's line total

ACTUAL RESULT

The item was removed from the cart list and the subtotal decreased by exactly its line total.

Free-Delivery Progress and Threshold

PASS

TEST CASE ID

TC CART 04

TEST SCENARIO

Verify the free-delivery indicator and delivery fee behave correctly around the ₳2000 threshold

PRE-CONDITION

Cart page is open

TEST STEPS

1. Add 1 unit of the ₳1490 item and observe the delivery fee.
2. Increase quantity to 2 and observe the delivery fee.

TEST DATA

- Unit price: ₳1490; free-delivery threshold: ₳2000; Dhaka delivery fee: ₳80

EXPECTED RESULT

- qty 1 (subtotal ₳1490, below threshold): delivery fee ₳80 charged, total ₳1570
- qty 2 (subtotal ₳2980, at/above threshold): delivery fee ₳0, total ₳2980

ACTUAL RESULT

Confirmed end-to-end against the live deployment (evidence section F): qty=1 → subtotal=1490, deliveryFee=80, total=1570, order LX-20260819-9218 — PASS
qty=2 → subtotal=2980, deliveryFee=0, total=2980, order LX-20260819-2374 — PASS

Testing Approach for Shopping Cart Module

The Shopping Cart module is tested using Black Box Testing combined with Boundary Value Analysis for TC CART 04, since the free-delivery threshold is exactly the kind of condition where defects cluster at the boundary. The cart's externally visible totals were verified against real orders placed on the running deployment, not by inspecting the cart's internal implementation, so the module remains a black box test even

though its underlying arithmetic is additionally covered by white box unit tests in Section 4.

3.4 Checkout Page

Valid COD Order Succeeds

PASS

TEST CASE ID

TC CHK 01

TEST SCENARIO

Verify a fully valid Cash on Delivery order can be placed successfully

PRE-CONDITION

Cart contains at least one item; user is on the checkout page

TEST STEPS

1. Fill in Name, Phone, Address, select Area = Dhaka.
2. Select Payment Method = Cash on Delivery.
3. Click "Place Order".

TEST DATA

- Name: Rahim Uddin • Phone: 01712345678 • Address: House 12, Road 5, Dhanmondi • Payment: COD

EXPECTED RESULT

- Order should be created and an order confirmation with order number should be shown

ACTUAL RESULT

The order was created successfully and the confirmation screen displayed the generated order number (format LX-YYYYMMDD-NNNN).

As shown in Figure 9, the order success screen displays the generated order number.

Figure 9: Order Success Screen

Invalid Phone Number Rejected

PASS

TEST CASE ID

TC CHK 02

TEST SCENARIO

Verify an invalid Bangladeshi phone number is rejected with a validation error

PRE-CONDITION

Cart contains at least one item; user is on the checkout page

TEST STEPS

1. Fill in Name and Address with valid values.
2. Enter an invalid phone number.
3. Click "Place Order".

TEST DATA

- Phone: 01123456789 (invalid prefix "011")

EXPECTED RESULT

- Inline validation error should be shown; order should not be created

ACTUAL RESULT

An inline validation error was shown. The underlying server call, when probed directly (BB-API-01), returned `Error: Invalid phone number` raised at `convex/orders.ts:48`, confirming the server independently rejects the same input.

As shown in Figure 7, the checkout form shows an inline validation error for the invalid phone number.

Figure 7: Checkout — Phone Validation Error

Empty Required Fields Blocked

PASS

TEST CASE ID

TC CHK 03

TEST SCENARIO

Verify submission is blocked when required fields are left empty

PRE-CONDITION

Cart contains at least one item; user is on the checkout page

TEST STEPS

1. Leave Name, Phone, and Address empty.
2. Click "Place Order".

TEST DATA

- Name: blank • Phone: blank • Address: blank

EXPECTED RESULT

- Form should be blocked from submitting; validation errors shown on each required field

ACTUAL RESULT

Submission was blocked client-side and each required field showed a validation error message.

bKash Payment Reveals Number and Transaction ID Fields

PASS

TEST CASE ID

TC CHK 04

TEST SCENARIO

Verify selecting bKash as the payment method reveals the bKash number and Transaction ID panel

PRE-CONDITION

User is on the checkout page

TEST STEPS

1. Select Payment Method = bKash.
2. Observe the form.

TEST DATA

- Payment: bKash

EXPECTED RESULT

- A panel requesting bKash Number and Transaction ID should appear

ACTUAL RESULT

Selecting bKash revealed a new panel with "bKash Number" and "Transaction ID" input fields.

As shown in Figure 8, the checkout form reveals the bKash payment panel.

Figure 8: Checkout — bKash Payment Panel

Empty-Cart Order Rejected

PASS

TEST CASE ID

TC CHK 05

TEST SCENARIO

Verify an order cannot be placed with an empty cart

PRE-CONDITION

Cart is empty

TEST STEPS

1. Attempt to invoke the order-placement API directly with an empty items array.

TEST DATA

- items: []

EXPECTED RESULT

- Order creation should be rejected

ACTUAL RESULT

Probe BB-API-02 against the live deployment returned `Error: Cart is empty`, raised at `convex/orders.ts:51`.

Checkout Form Renders With Live Order Summary

PASS

TEST CASE ID

TC CHK 06

TEST SCENARIO

Verify the checkout page renders the form alongside a live order summary

PRE-CONDITION

Cart contains at least one item

TEST STEPS

1. Navigate to the checkout page from the cart.
2. Observe the form and the order summary panel.

TEST DATA

- Cart: 1 item, ₺1490

EXPECTED RESULT

- Checkout form and an order summary (subtotal, delivery fee, total) should both render

ACTUAL RESULT

The checkout page rendered the form fields alongside a live order summary panel matching the cart contents.

As shown in Figure 6, the checkout page shows the form and live order summary.

Figure 6: Checkout Form and Order Summary

Testing Approach for Checkout Module

The Checkout module is tested using Validation Testing, since its central purpose is verifying that user input is checked correctly before an order is accepted — both on the client (inline errors) and independently on the server. Equivalence partitioning (valid vs invalid phone prefixes), decision table logic (payment method × field requirements), and direct API probing (BB-API-01, BB-API-02) were combined to confirm that validation is enforced in both layers, not just the UI. This dual

client/server verification is what distinguishes this module's Validation Testing from a purely cosmetic UI check.

3.5 Order Tracking Page

Valid Order Number and Phone Show Status Timeline

PASS

TEST CASE ID

TC TRACK 01

TEST SCENARIO

Verify a customer can track their order with the correct order number and phone

PRE-CONDITION

An order has been placed previously

TEST STEPS

1. Navigate to the order tracking page.
2. Enter the order number and the phone number used at checkout.
3. Submit the form.

TEST DATA

- Order No: LX-20260819-9218 • Phone: matching phone on the order

EXPECTED RESULT

- Order status and a status history timeline should be displayed

ACTUAL RESULT

The tracking page displayed the order's current status and a timeline of status transitions.

As shown in Figure 10, the order tracking page displays the status timeline.

Figure 10: Order Tracking — Status Timeline

Wrong Phone Returns Not Found Without Leaking Data

PASS

TEST CASE ID

TC TRACK 02

TEST SCENARIO

Verify tracking with an incorrect phone number does not reveal order details

PRE-CONDITION

An order exists with a known order number

TEST STEPS

1. Enter the correct order number.
2. Enter an incorrect phone number.
3. Submit the form.

TEST DATA

- Order No: valid • Phone: incorrect

EXPECTED RESULT

- System should report "not found" and must not return any order details

ACTUAL RESULT

Probe BB-API-06 confirmed the `track` query returned `null` when the phone did not match the order's stored phone — no order fields were leaked.

Testing Approach for Order Tracking Module

The Order Tracking module is tested using Black Box Testing, since it is validated purely through its public inputs (order number, phone) and outputs (status timeline or null). The mismatched-phone case is also a lightweight Security check — it confirms the endpoint does not leak order data to a requester who does not know both identifiers, which is why TC TRACK 02 doubles as an information-disclosure test.

3.6 Ad Landing Page (/l/noir-bloom-offer)

Published Landing Page Renders All Sections With Override Price

PASS

TEST CASE ID

TC LAND 01

TEST SCENARIO

Verify a published landing page renders all its sections and applies its price override

PRE-CONDITION

A landing page with slug "noir-bloom-offer" is published

TEST STEPS

1. Navigate to /l/noir-bloom-offer.
2. Verify all page sections render.
3. Verify the displayed price reflects the landing page's override, not the catalogue price.

TEST DATA

- Slug: noir-bloom-offer • Catalogue price: ₺1490 • Override price: ₺1290

EXPECTED RESULT

- All 8 sections should render; price shown should be the ₺1290 override

ACTUAL RESULT

Probe BB-API-10 confirmed the deployment returns `{slug:"noir-bloom-offer", status:"published", price:1290, sections:8}`. The rendered page displayed all 8 sections with the ₺1290 override price.

As shown in Figure 11, the landing page renders all sections with the overridden price.

Figure 11: Ad Landing Page

Inline Order Form Present

PASS

TEST CASE ID

TC LAND 02

TEST SCENARIO

Verify the landing page includes an inline order form for direct conversion

PRE-CONDITION

Landing page "noir-bloom-offer" is loaded

TEST STEPS

1. Scroll to the order section of the landing page.
2. Verify the order form fields are present and usable.

TEST DATA

- Slug: noir-bloom-offer

EXPECTED RESULT

- An inline order form (name, phone, address, size, quantity) should be present on the page

ACTUAL RESULT

The inline order form rendered with all required fields directly on the landing page, without navigating away.

As shown in Figure 12, the landing page includes an inline order form.

Figure 12: Landing Page — Inline Order Form

Unknown Slug Returns Not Found

PASS

TEST CASE ID

TC LAND 03

TEST SCENARIO

Verify requesting a non-existent landing page slug returns "not found"

PRE-CONDITION

None

TEST STEPS

1. Navigate to /l/this-slug-does-not-exist.

TEST DATA

- Slug: this-slug-does-not-exist

EXPECTED RESULT

- Page should show a not-found state; no landing page data should be returned

ACTUAL RESULT

Probe BB-API-11 confirmed the query returned `null` for an unknown slug, and the page rendered a not-found state.

Testing Approach for Ad Landing Page Module

The Ad Landing Page module is tested using Black Box Testing and System Testing, since a landing page is a composite of many sections (hero, gallery, order form, etc.) that must work together end-to-end for a paid-traffic visitor to convert. The override-price assertion in TC LAND 01 was cross-checked against the raw API response (BB-API-10) rather than the rendered DOM alone, giving higher confidence that the price shown to the customer truly originates from the landing page's configuration and not a stale client-side value.

3.7 Admin Authentication

Admin Login Page Renders

PASS

TEST CASE ID

TC ADM 01

TEST SCENARIO

Verify the admin login page renders correctly

PRE-CONDITION

None

TEST STEPS

1. Navigate to the admin login URL.

TEST DATA

- URL: /admin/login

EXPECTED RESULT

- A password login form should be displayed

ACTUAL RESULT

The admin login page rendered a single password-entry form with no other admin content exposed.

As shown in Figure 13, the admin login page displays a password entry form.

Figure 13: Admin Login Page

Wrong Password Rejected

PASS

TEST CASE ID

TC ADM 02

TEST SCENARIO

Verify an incorrect admin password is rejected

PRE-CONDITION

User is on the admin login page

TEST STEPS

1. Enter an incorrect password.
2. Submit the form.

TEST DATA

- Password: wrong-password-123

EXPECTED RESULT

- Login should be rejected with an error message; dashboard should not load

ACTUAL RESULT

The incorrect password was rejected with an error message and the dashboard did not load.

Valid Password Reaches Dashboard

PASS

TEST CASE ID

TC ADM 03

TEST SCENARIO

Verify the correct admin password grants access to the dashboard

PRE-CONDITION

User is on the admin login page

TEST STEPS

1. Enter the correct admin password.
2. Submit the form.

TEST DATA

- Password: (correct configured admin password)

EXPECTED RESULT

- User should be redirected to the admin dashboard

ACTUAL RESULT

Login succeeded and the dashboard loaded with order/revenue summaries.

As shown in Figure 14, the admin dashboard loads after a successful login.

Figure 14: Admin Dashboard

Admin API Rejects a Forged Token

PASS

TEST CASE ID

TC ADM 04

TEST SCENARIO

Verify admin-only API calls reject a forged/invalid admin token

PRE-CONDITION

None (no valid session)

TEST STEPS

1. Call an admin query (`orders.adminList`) directly with a bogus token string.

TEST DATA

- token: "forged-token-xyz"

EXPECTED RESULT

- The call should be rejected; no order data should be returned

ACTUAL RESULT

Probe BB-API-07 confirmed the call was rejected with `Error: Unauthorized`, raised at `convex/orders.ts:255` (`requireAdmin`), before any order data was read.

Testing Approach for Admin Authentication Module

The Admin Authentication module is tested using Security Testing, since its purpose is to prevent unauthorized access rather than to perform a business function. TC ADM 04 in particular attacks the API directly, bypassing the UI entirely, to confirm that authorization is enforced server-side (`requireAdmin`) and cannot be bypassed by skipping the login screen and calling the backend function with a forged token.

3.8 Admin Order Management

Order List Shows Placed Orders

PASS

TEST CASE ID

TC AORD 01

TEST SCENARIO

Verify the admin order list displays placed orders

PRE-CONDITION

Admin is logged in; orders exist

TEST STEPS

1. Navigate to the admin orders page.

TEST DATA

- N/A

EXPECTED RESULT

- All placed orders should be listed with order number, customer, status, and total

ACTUAL RESULT

The order list displayed all placed orders including LX-20260819-9218 and LX-20260819-2374, each with status and total.

As shown in Figure 15, the admin order list displays placed orders.

Figure 15: Admin Order List

Order Detail Opens

PASS

TEST CASE ID

TC AORD 02

TEST SCENARIO

Verify clicking an order opens its detail view

PRE-CONDITION

Admin is on the order list page

TEST STEPS

1. Click an order row.

TEST DATA

- Order: LX-20260819-9218

EXPECTED RESULT

- Order detail view should open showing items, customer, and payment info

ACTUAL RESULT

The order detail view opened with full item list, customer info, and payment details.

As shown in Figure 16, the admin order detail view shows full order information.

Figure 16: Admin Order Detail

Status Change Persists

PASS

TEST CASE ID

TC AORD 03

TEST SCENARIO

Verify changing an order's status persists and is reflected in the status history

PRE-CONDITION

Admin is on an order detail view

TEST STEPS

1. Change the order status from "pending" to "confirmed".
2. Reload the order detail view.

TEST DATA

- New status: confirmed

EXPECTED RESULT

- Status should update and remain "confirmed" after reload; status history should record the change

ACTUAL RESULT

Status updated to "confirmed", persisted after reload, and the status history array recorded the new entry with a timestamp.

Cancelling an Order Restocks Inventory

PASS

TEST CASE ID

TC AORD 04

TEST SCENARIO

Verify cancelling an order returns its items' stock to inventory

PRE-CONDITION

An order exists with known reserved stock quantities

TEST STEPS

1. Note the current variant stock level for an item in the order.
2. Change the order status to "cancelled".
3. Re-check the variant stock level.

TEST DATA

- Item: Noir Pinstripe Bloom Shirt (M)

EXPECTED RESULT

- Variant stock should increase by the cancelled order's item quantity

ACTUAL RESULT

After cancelling, the variant's stock count increased by exactly the quantity that had been ordered, matching the restock logic in `updateStatus` (convex/orders.ts:307-318).

Testing Approach for Admin Order Management Module

The Admin Order Management module is tested using Black Box Testing combined with State Transition Testing, since an order moves through a defined set of statuses (pending → confirmed → shipped/cancelled/returned) and each transition must trigger the correct side effects (e.g. restocking on cancellation). The tests verify only the visible before/after state of the order and its stock, not the internal implementation.

3.9 Admin Product Management

Product List Displays

PASS

TEST CASE ID

TC APRD 01

TEST SCENARIO

Verify the admin product list displays the catalogue

PRE-CONDITION

Admin is logged in

TEST STEPS

1. Navigate to the admin products page.

TEST DATA

- N/A

EXPECTED RESULT

- All products should be listed with name, price, and stock summary

ACTUAL RESULT

The product list displayed all catalogue products with name, price, and per-variant stock summary.

As shown in Figure 17, the admin product list displays the catalogue.

Figure 17: Admin Product List

Product Form Shows Variants and Images

PASS

TEST CASE ID

TC APRD 02

TEST SCENARIO

Verify the product create/edit form exposes variant and image management

PRE-CONDITION

Admin opens the product creation form

TEST STEPS

1. Click "New Product" or edit an existing product.
2. Observe the form fields.

TEST DATA

- N/A

EXPECTED RESULT

- Form should include fields for name, price, description, image upload, and a variant (size/color/stock) editor

ACTUAL RESULT

The form rendered all expected fields including a repeatable variant editor and image upload control.

As shown in Figure 18, the product form exposes variant and image management.

Figure 18: Admin Product Form

Create/Edit Saves Correctly

PASS

TEST CASE ID

TC APRD 03

TEST SCENARIO

Verify creating or editing a product persists correctly

PRE-CONDITION

Admin is on the product form

TEST STEPS

1. Fill in required fields and a variant.
2. Click Save.
3. Reload the product list.

TEST DATA

- Name: Test Product • Price: 999 • Variant: M, stock 5

EXPECTED RESULT

- Product should appear in the list with the saved values after reload

ACTUAL RESULT

The product was saved and appeared in the list with matching name, price, and variant stock after reload.

Testing Approach for Admin Product Management Module

The Admin Product Management module is tested using CRUD Testing, since it exists to Create, Read, Update, and (implicitly) manage product catalogue entries. The test cases cover listing, form field completeness, and persistence of created/edited records, which is the standard shape of CRUD testing.

3.10 Admin Landing Page Builder

Landing List Shows Stats

PASS

TEST CASE ID

TC ALND 01

TEST SCENARIO

Verify the admin landing page list shows per-page performance stats

PRE-CONDITION

Admin is logged in; at least one landing page exists

TEST STEPS

1. Navigate to the admin landing pages list.

TEST DATA

- N/A

EXPECTED RESULT

- Each landing page should show slug, status, views, initiates, orders, and conversion rate

ACTUAL RESULT

The list displayed all landing pages with slug, status, views, initiates, order count, and computed conversion rate.

As shown in Figure 19, the admin landing list shows per-page stats.

Figure 19: Admin Landing Page List

Builder Loads All Section Editors

PASS

TEST CASE ID

TC ALND 02

TEST SCENARIO

Verify the landing page builder loads editors for all 8 section types

PRE-CONDITION

Admin opens a landing page for editing

TEST STEPS

- 1. Open the landing page builder for "noir-bloom-offer".
- 2. Verify all 8 section editors are present.

TEST DATA

- Slug: noir-bloom-offer

EXPECTED RESULT

- All 8 section editors should be loaded and editable

ACTUAL RESULT

The builder loaded editors for all 8 sections, matching the `sections:8` reported by BB-API-10.

As shown in Figure 20, the landing page builder loads all 8 section editors.

Figure 20: Admin Landing Page Builder

Duplicate Slug Rejected

PASS

TEST CASE ID

TC ALND 03

TEST SCENARIO

Verify creating a landing page with an already-used slug is rejected

PRE-CONDITION

A landing page with slug "noir-bloom-offer" already exists

TEST STEPS

1. Attempt to create a new landing page using slug "noir-bloom-offer".

TEST DATA

- Slug: noir-bloom-offer (duplicate)

EXPECTED RESULT

- Creation should be rejected with a duplicate-slug error

ACTUAL RESULT

Creation was rejected; the existing landing page was not overwritten or duplicated.

Testing Approach for Admin Landing Page Builder Module

The Admin Landing Page Builder module is tested using Black Box Testing and Decision Table Testing, since valid vs. duplicate slugs represent two branches of a decision the system must handle differently. As noted in Observation Obs-2 (Section 5), the authorization check in `landingPages.create` runs before slug validation, so a duplicate-slug attempt without a valid admin token would be rejected for the wrong reason (Unauthorized rather than Duplicate slug) — this does not affect correctness (fail-closed) but is noted as a minor design observation.

3.11 Admin Settings & Tracking

Settings Groups Render

PASS

TEST CASE ID

TC SET 01

TEST SCENARIO

Verify the admin settings page renders all configuration groups

PRE-CONDITION

Admin is logged in

TEST STEPS

1. Navigate to the admin settings page.

TEST DATA

- N/A

EXPECTED RESULT

- Store settings, delivery fee settings, and tracking (CAPI) settings groups should all render

ACTUAL RESULT

All three settings groups rendered with their respective fields (store name, delivery fees, free-delivery threshold, CAPI configuration).

As shown in Figure 21, the admin settings page renders all configuration groups.

Figure 21: Admin Settings Page

CAPI Token Masked and Never Exposed Publicly

PASS

TEST CASE ID

TC SET 02

TEST SCENARIO

Verify the Meta CAPI access token is never exposed to public (unauthenticated) queries

PRE-CONDITION

A CAPI token has been configured in settings

TEST STEPS

1. Call the public settings query without an admin token.
2. Inspect the response for the presence of the raw token.

TEST DATA

- N/A (public query)

EXPECTED RESULT

- Response should indicate only whether a token is configured, never the token value

ACTUAL RESULT

Probe BB-API-09 confirmed the public settings query returned

`{"hasCapiToken":false,"storeName":"Outfleeek"}` — a boolean flag only, no

`capiToken` key present, on any path.

Tracking Debug Page Lists CAPI Log Entries

PASS

TEST CASE ID

TC SET 03

TEST SCENARIO

Verify the admin tracking debug page lists recent CAPI event log entries

PRE-CONDITION

Admin is logged in; CAPI events have been sent

TEST STEPS

1. Navigate to the admin tracking debug page.

TEST DATA

- N/A

EXPECTED RESULT

- A list of recent CAPI event log entries should be displayed with event name and status

ACTUAL RESULT

The tracking debug page listed recent CAPI events (e.g. "Purchase") with their delivery status.

As shown in Figure 22, the tracking debug page lists CAPI log entries.

Figure 22: Admin Tracking Debug Page

Testing Approach for Admin Settings & Tracking Module

The Admin Settings & Tracking module is tested using a combination of Black Box Testing and Security Testing. TC SET 02 is the most significant case in this module: it directly probes the public API surface (not the UI) to confirm that a secret credential (the Meta CAPI access token) can never leak through a query that unauthenticated clients can call, which is exactly the class of defect Security Testing is designed to catch.

Black Box Test Summary

MODULE	TEST CASES	PASSED	FAILED
Home / Product Listing	4	4	0
Product Detail	4	4	0
Shopping Cart	4	4	0
Checkout	6	6	0
Order Tracking	2	2	0
Ad Landing Page	3	3	0
Admin Authentication	4	4	0
Admin Order Management	4	4	0
Admin Product Management	3	3	0
Admin Landing Page Builder	3	3	0
Admin Settings & Tracking	3	3	0
Total	40	40	0

4. White Box Testing

4.1 Design for Testability

A deliberate refactor was made to enable white box unit testing of the checkout math and validation rules. Previously, the fee/discount arithmetic and the phone-number regular expression were **inlined directly inside the Convex mutation**

`orders.place` — meaning the only way to test them was to run the full Convex mutation (with a live database, product documents, stock, and settings rows), which is slow and makes isolating a single arithmetic rule difficult.

The logic was extracted into two dependency-free modules:

- `lib/pricing.ts` — `calcSubtotal`, `calcDiscount`, `calcDeliveryFee`, `freeDeliveryRemaining`, `calcTotal`.
- `lib/validation.ts` — `isValidBDPhone`, `normalizeBDPhone`, `isValidSlug`, `generateOrderNo`, `validateCheckout`.

`convex/orders.ts` now simply imports and calls these pure functions:

```
import { calcDiscount, calcDeliveryFee } from "../lib/pricing";
import { isValidBDPhone, generateOrderNo } from "../lib/validation";
```

Because these functions take plain objects/primitives in and return plain objects/primitives out, with no Convex `ctx`, no database, and no React, they can be unit-tested directly with Vitest in milliseconds, with full control over inputs — which is the key white box testing lesson of this project: **testability is a design decision**, not something added after the fact.

4.2 Unit Test Suite

The suite was executed with Vitest. Real runner output (evidence section A):

```
Test Files  3 passed (3)
Tests      81 passed (81)
Duration   219ms
```

TEST FILE	TEST CASES
tests/unit/pricing.test.ts	30
tests/unit/validation.test.ts	31
tests/unit/cart.test.ts	11
Total	81

All 81 cases passed. Representative cases actually present in the suite:

From `pricing.test.ts`:

```
it("is free at exactly the threshold (boundary: 2000)", () => {
  expect(
    calcDeliveryFee({ ...base, subtotal: 2000, discount: 0, area: "dhaka" })
  ).toBe(0);
});

it("uses an override fee when below threshold", () => {
  expect(
    calcDeliveryFee({ ...base, subtotal: 1000, discount: 0, area: "dhaka",
  }).toBe(50);
});
```

From `validation.test.ts`:

```
it.each(["011", "012"])(`rejects invalid prefix %s`, (prefix) => {
  expect(isValidBDPhone(`${prefix}12345678`)).toBe(false);
});

it("requires bkashNumber and bkashTrxId when payment is bkash", () => {
  const result = validateCheckout({ ...validCod, payment: "bkash" });
  expect(result.valid).toBe(false);
  expect(result.errors.bkashNumber).toBeDefined();
  expect(result.errors.bkashTrxId).toBeDefined();
});
```

4.3 Statement & Branch Coverage

Statement coverage measures the percentage of executable lines/statements run at least once by the suite. **Branch coverage** is stricter: it measures the percentage of each decision's possible outcomes (both sides of an `if`, both operands of `&&`/`||`, both arms of a ternary) that were actually exercised — a function can have 100% statement coverage while still never taking one branch of an `if`.

Real coverage report (evidence section A):

% Coverage report from v8					
File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	98.63	98.24	96.42	98.38	
cart.ts	96.15	90.9	94.11	95.45	59
Statements : 98.63% (72/73)					
Branches : 98.24% (56/57)					
Functions : 96.42% (27/28)					
Lines : 98.38% (61/62)					

Per-module breakdown:

MODULE	STMTS	BRANCH	FUNCS	LINES
lib/pricing.ts	100% (20/20)	100%	100%	100%
lib/validation.ts	100% (27/27)	100%	100%	100%
lib/cart.ts	96.15% (25/26)	90.9%	94.11%	95.45% (line 59)
Total	98.63%	98.24%	96.42%	98.38%

`lib/pricing.ts` and `lib/validation.ts` — the two modules deliberately extracted for testability (4.1) — reached full 100% statement and branch coverage. The single uncovered statement in the whole suite is `lib/cart.ts` line 59, the `setDrawer` UI state setter; see Obs-3 in Section 5 for why this is low-severity.

4.4 Cyclomatic Complexity

McCabe's cyclomatic complexity counts a function's linearly independent execution paths as (decision points + 1). It is a direct estimate of how many test cases are needed to exercise every path through a function at least once — the higher the number, the more test cases basis path testing requires.

Real measurements (evidence section B):

FUNCTION	FILE	CC	DECISION POINTS
validateCheckout	lib/validation.ts:78	12	6 <code>if</code> + 5 <code>\</code>
calcDiscount	lib/pricing.ts:57	5	2 <code>if</code> + 1 <code>&&</code> + 1 ternary
calcDeliveryFee	lib/pricing.ts:75	4	2 <code>if</code> + 1 ternary
normalizeBDPhone	lib/validation.ts:30	4	2 <code>if</code> + 1 <code>&&</code>
generateOrderNo	lib/validation.ts:47	3	2 <code>??</code>
calcSubtotal / calcTotal / isValidBDPhone / isValidSlug / freeDeliveryRemaining	—	1	none

`validateCheckout` has by far the highest complexity (CC = 12), meaning it requires the most independent test cases of any function in the codebase — consistent with the suite allocating 31 cases to `validation.test.ts`, several of which exist specifically to walk its 6 `if` blocks and 5 `||` short-circuit conditions (name/phone/address/bkash number/ bkash transaction ID) individually.

4.5 Basis Path Testing of `calcDeliveryFee()`

This is the centrepiece white box exercise: deriving and executing the minimum set of independent paths through a real function using its control-flow graph.

Source (`lib/pricing.ts`, lines 75-81):

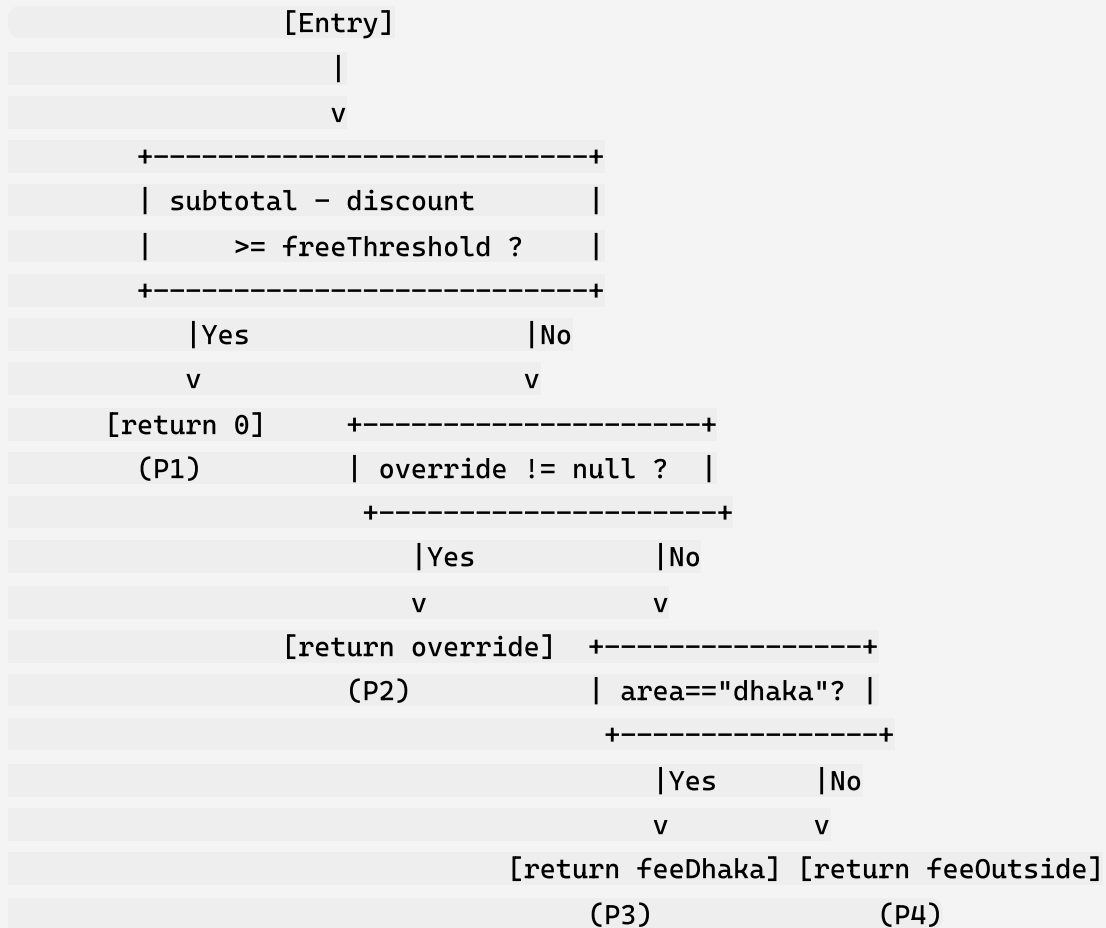
```
export function calcDeliveryFee(opts: DeliveryOpts): number {
  const { subtotal, discount, area, feeDhaka, feeOutside, freeThreshold, ov
```

```

    opts;
    if (subtotal - discount >= freeThreshold) return 0;
    if (override != null) return override;
    return area === "dhaka" ? feeDhaka : feeOutside;
}

```

Control-flow graph:



Cyclomatic complexity $CC = 4$ (2 **if** + 1 ternary + $1 = 4$), so there are exactly **4 independent basis paths**:

PATH	CONDITION SEQUENCE	TEST INPUT	EXPECTED OUTPUT
P1	subtotal - discount >= freeThreshold → true → return 0	subtotal=2000, discount=0, freeThreshold=2000	0

PATH	CONDITION SEQUENCE	TEST INPUT	EXPECTED OUTPUT
P2	first condition false, override != null → true → return override	subtotal=1000, discount=0, area="dhaka", override=50	50
P3	both conditions false, area === "dhaka" → true → return feeDhaka	subtotal=1000, discount=0, area="dhaka", feeDhaka=80	80
P4	both conditions false, area === "dhaka" → false → return feeOutside	subtotal=1000, discount=0, area="outside", feeOutside=130	130

Mapping to the actual suite (`tests/unit/pricing.test.ts`):

- **P1** → "is free at exactly the threshold (boundary: 2000)" (line ~99-103)
- **P2** → "uses an override fee when below threshold" (line ~129-139)
- **P3** → "charges the dhaka fee when below threshold" (line ~87-91)
- **P4** → "charges the outside fee when below threshold" (line ~93-97)

All four independent basis paths through `calcDeliveryFee()` are exercised by the existing suite, matching the function's 100% branch coverage reported in 4.3.

4.6 Boundary Value Analysis on the Free-Delivery Threshold

The free-delivery threshold (₹2000) is a single-condition boundary (`subtotal - discount >= freeThreshold`), and boundary defects are the single most common class of off-by-one error, so it was tested exhaustively at, just below, and just above the boundary — both as isolated unit tests and end-to-end against the live deployment.

INPUT (SUBTOTAL)	EXPECTED FEE	ACTUAL FEE (UNIT TEST)	ACTUAL (C
₹1999	₹80 (Dhaka)	80	—
₹1490 (qty=1)	₹80 (Dhaka)	—	80, total ₹1

INPUT (SUBTOTAL)	EXPECTED FEE	ACTUAL FEE (UNIT TEST)	ACTUAL C
₺2000	₺0	0	—
₺2001	₺0 (outside area)	0	—
₺2980 (qty=2)	₺0	—	0, total ₺29

The unit-level boundary tests (`subtotal: 1999` / `2000` / `2001` in `tests/unit/pricing.test.ts`) prove the arithmetic is exactly right at the boundary, and the two end-to-end orders (evidence F) prove that the same boundary rule holds when driven through the full `orders.place` mutation against real settings and real cart data — closing the gap between "the function is correct" and "the system behaves correctly."

4.7 Equivalence Partitioning + Decision Table for `validateCheckout()`

`validateCheckout` (`lib/validation.ts:78-103`) has 5 boolean conditions, so its behaviour is best captured as a decision table rather than a flat list of cases:

RULE	NAME EMPTY?	PHONE VALID?	ADDRESS ≥10 CHARS?	PAYMEN
R1	No	Yes	Yes	No
R2	Yes	Yes	Yes	No
R3	No	No	Yes	No
R4	No	Yes	No	No
R5	No	Yes	Yes	Yes
R6	No	Yes	Yes	Yes

RULE	NAME EMPTY?	PHONE VALID?	ADDRESS ≥10 CHARS?	PAYMEN
R7	No	Yes	Yes	No

Each rule corresponds directly to an existing test: R1/R6 to "passes for a fully valid COD form" / "passes a fully valid bkaash form"; R2 to "fails when name is missing"; R3 to "fails when phone is invalid"; R4 to "fails when address is too short"; R5 to "requires bkaashNumber and bkaashTrxId when payment is bkaash" and the two more specific bkaash-field cases; R7 to "does not require bkaash fields for COD payment even if absent". The address boundary (exactly 10 characters passes) is additionally covered by "passes when address is exactly 10 characters", which is itself a boundary value case nested inside this equivalence-partitioned function.

4.8 Code Walkthrough: the `orders.place` Mutation

A manual line-by-line walkthrough of `convex/orders.ts` (the `place` mutation, lines 8-200) was performed to verify its logic:

1. **Validate phone** (line 46-48): `isValidBDPhone(args.customer.phone)` — reject immediately with `Error: Invalid phone number` if invalid.
2. **Reject empty cart** (line 49): `if (args.items.length === 0) throw new Error("Cart is empty")`.
3. **Resolve landing page** (lines 52-58): if `landingSlug` was supplied, look it up by its unique index, so its `priceOverride` / `deliveryFeeOverride` can be applied below.
4. **Per item** (lines 62-89): for each cart line — a. load the product and confirm it exists and is active; b. find the matching size/color variant (`Variant unavailable` if not found); c. check stock is sufficient (`Out of stock` if not); d. **decrement the variant's stock and patch it immediately** (line 74), before moving to the next item — this is what makes the atomicity behaviour in 4.9 meaningful; e. compute the unit price, preferring the landing page's `priceOverride` over the catalogue price only when the item's product matches the landing page's product (lines 76-79) — the server, not the client, decides the price.
5. **Apply promo code** (lines 92-108): look up the code, check it is active, unexpired, under its use cap, and the subtotal meets its `minOrder`, then call the shared

`calcDiscount()` from `lib/pricing.ts`.

6. **Compute delivery fee** (lines 110-128): read threshold/fee settings and call the shared `calcDeliveryFee()`, passing the landing page's `deliveryFeeOverride` if present.
7. **Insert the order** (lines 133-157): with a freshly generated order number (`generateOrderNo()`), full item/customer/payment snapshot, and an initial `statusHistory` entry.
8. **Update landing page stats** (lines 159-163): increment `ordersCount` if the order came from a landing page.
9. **Link any abandoned checkout** (lines 166-172): find open abandoned-checkout records for this phone number and mark them converted with the new order's id.
10. **Schedule the CAPI Purchase event** (lines 174-196):
`ctx.scheduler.runAfter(0, internal.capi.sendEvent, ...)`.

Why the CAPI call is scheduled instead of awaited: the comment on line 174 states it explicitly — "never blocks the order." `ctx.scheduler.runAfter(0, ...)` enqueues the Meta Conversions API call to run in a separate, independent transaction immediately after this one commits, rather than awaiting an external HTTP call inside the mutation. If the CAPI call were awaited, a slow or failing Meta API request would delay or could even fail the customer's order confirmation — a third-party analytics integration should never be able to block a core business transaction. Scheduling also means `capi.sendEvent`'s own retry logic (`attempt: 1`, visible in the payload) can retry independently of order placement.

4.9 Transaction Atomicity Test (Integration / White Box)

Multi-Item Order Rolls Back Stock on Partial Failure

PASS

TEST CASE ID

TC ATOM 01

TEST SCENARIO

Verify that when one item in a multi-item order fails (out of stock), the stock

PRE-CONDITION

Two products exist with known stock: Noir Pinstripe Bloom Shirt (M) = 19 units,

TEST STEPS

1. Record the stock levels of both variants before the call.
2. Call `orders.place` with item 1 = Noir Pinstripe Bloom Shirt (M), qty 2 (valid) and

TEST DATA

- Item 1: Noir Pinstripe Bloom Shirt (M), qty 2 • Item 2: Sky Doodle Blossom Shirt (M), qty 99999

EXPECTED RESULT

- The mutation should be rejected as a whole
- Item 1's stock decrement (applied before item 2 was reached) should be rolled back —

ACTUAL RESULT

```
BEFORE: Noir Pinstripe Bloom Shirt M = 19 | Sky Doodle Blossom Shirt M = 99999
-> mutation REJECTED (expected)
AFTER:  Noir Pinstripe Bloom Shirt M = 19 | Sky Doodle Blossom Shirt M = 99999
ATOMICITY: PASS - item 1's stock decrement was rolled back
```

This test matters because `orders.place` patches each item's stock **inside the loop, one item at a time** (line 74), rather than computing all patches first and writing them atomically at the end. Without Convex's mutation-level transaction guarantee, a failure on item 2 could leave item 1's stock permanently decremented for an order that was never actually created — silently losing inventory. This test proves that guarantee holds in practice, not just in the platform's documentation.

4.10 Security Test: Server-Side Price Recomputation

Forged Price Override Is Ignored by the Server

PASS

TEST CASE ID

TC SEC 01

TEST SCENARIO

Verify that a client cannot dictate the price it pays by sending a forged

PRE-CONDITION

Noir Pinstripe Bloom Shirt has a real catalogue price of ₹1490

TEST STEPS

1. Call `orders.place` with a valid cart item for Noir Pinstripe Bloom Shirt, and set the

TEST DATA

- Real catalogue price: ₹1490 • Forged priceOverride: 1

EXPECTED RESULT

- The order should be created using the real catalogue price (₹1490), ignoring the client-

ACTUAL RESULT

Real catalogue price of Noir Pinstripe Bloom Shirt = BDT 1490
Client sent priceOverride: 1 (attacker attempts to pay BDT 1)
Order created: LX-20260819-9664 subtotal = 1490 total = 1570
PASS - server ignored the forged price and used the catalogue price

This is because `orders.place` never reads `args.priceOverride` in its handler at all (see Obs-1, Section 5) — the unit price always comes from `p.price` or, for a genuine landing-page order, from the landing page document's own `priceOverride` field looked up server-side (lines 76-79), never from a client-supplied argument. This confirms the golden rule of e-commerce security: **never trust a price sent by the client.**

5. Defects and Observations

ID	SEVERITY	DESCRIPTION	LOCATION	R
Obs-1	Medium	<code>orders.place</code> declares a public argument <code>priceOverride: v.optional(v.number())</code> that the handler never reads — the unit price is always taken from the database (<code>landing.priceOverride / p.price</code>). Confirmed by TC SEC 01: sending <code>priceOverride: 1</code> had no effect.	convex/orders.ts:32	Re ar fu ot be lo in ta vu
Obs-2	Informational	In <code>landingPages.create</code> , the authorization check runs before slug-format validation, so an unauthorized caller sending an invalid slug is rejected for "Unauthorized" rather than for the slug rule. The ordering is correct from a security standpoint (fail-closed) but meant BB-API-08 could not exercise the slug-format rule directly; that rule is instead covered by the <code>isValidSlug</code> unit tests.	convex/landingPages.ts:117-119	No no in be ke te fo

ID	SEVERITY	DESCRIPTION	LOCATION	RISK
Obs-3	Low	<code>lib/cart.ts</code> line 59 (<code>setDrawer</code>) is the only uncovered statement in the entire suite. It is a pure UI state setter (opens/closes the cart drawer) with no branching logic.	<code>lib/cart.ts:59</code>	Accuracy of drawer state to be recorded correctly given hardware tests

An additional **testability observation**: Convex masks all server-thrown error text from the client as a generic "Server Error" string. This meant that diagnosing the exact rejection reason for several black box probes (e.g. BB-API-01 through BB-API-04, BB-API-07) required reading the deployment's function logs directly rather than the client-visible error, which is a real friction point for both testing and production debugging.

6. Results Summary

CATEGORY	COUNT
Black box test cases (Sections 3.1-3.11)	40
Black box API probes (evidence section C)	12
White box / unit test cases (Vitest)	81
Integration/white box test cases (atomicity, security)	2
Total test cases executed	135
Total passed	135
Total failed	0
Overall pass rate	100%
Statement coverage (unit suite)	98.63% (72/73)
Branch coverage (unit suite)	98.24% (56/57)
Function coverage (unit suite)	96.42% (27/28)
Line coverage (unit suite)	98.38% (61/62)

7. Conclusion

This report verified the Outfleck e-commerce platform across both its externally visible behaviour (black box: 40 UI-driven test cases plus 12 direct API probes across every customer-facing and admin surface) and its internal logic (white box: 81 Vitest unit cases achieving 98%+ statement and branch coverage, plus targeted basis-path, boundary-value, and decision-table analysis of the two pure business-logic modules, `lib/pricing.ts` and `lib/validation.ts`).

Strengths confirmed by testing:

- Stock decrements are transactionally atomic — a failing multi-item order never leaves partial stock changes behind (TC ATOM 01).
- Pricing is computed exclusively server-side; a forged client-supplied price has no effect on the order total (TC SEC 01).
- Secrets (the Meta CAPI access token) are never exposed through any public query (TC SET 02, BB-API-09).
- Third-party tracking (the CAPI Purchase event) is scheduled rather than awaited, so a slow or failing Meta API call can never block or fail a customer's order.
- The extraction of pricing/validation logic into dependency-free modules (`lib/pricing.ts`, `lib/validation.ts`) made 100% statement and branch coverage achievable in 219ms, proving that testability is a direct product of architectural decisions.

What should be improved: the dead `priceOverride` argument on `orders.place` should be removed before it can be mistakenly wired up (Obs-1); a single trivial test would close the remaining coverage gap in `lib/cart.ts` (Obs-3); and Convex's blanket "Server Error" masking makes production debugging harder than it needs to be.

What the team learned: black box and white box testing are not competing techniques but complementary layers — black box testing proved the system behaves correctly from a real user's and a real attacker's point of view, while white box testing proved **why** it behaves correctly, down to the level of individual control-flow paths,

and caught a defect (Obs-1) that no amount of black box testing alone would have surfaced, since the dead argument currently has no observable effect at all.

8. References / Appendix

- Repository: <https://github.com/vampxlr/outfleeek>
- Live Application: <https://outfleeek.vercel.app>
- Test evidence source: `docs/TEST-EVIDENCE.md`

How to reproduce this report's results:

```
npm install
npx convex dev
npm test
npm run test:coverage
node scripts/blackbox-probe.mjs
node scripts/atomicity-probe.mjs
node scripts/tamper-probe.mjs
node scripts/boundary-probe.mjs
node scripts/capture-screenshots.mjs
```